Engineering Chemistry · Unit_04 · Functional materials and green chemistry

DRAFT — UE26CY151A · Unit_04 · Functional materials and green chemistry · Synthesis and properties of epoxy resin

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Synthesis and properties of epoxy resin

1 Synthesis and properties of epoxy resin

Study focus: Synthesis and properties of epoxy resin. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000098 · EV0000099

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module 7 on functional materials on polymers, and today we are going to study about adhesives and specifically about epoxy resin.

Key details

- Only when it solidifies then the two surfaces get locked to each other.
- The examples would be phenol formaldehydrasen, urea formal dehydration, and epoxytes.
- How do these actually work?
- After that, the most important step is to add a curing agent.
- What does this curing agent do?
- It is going to trigger cross-linking into the adhesive.
- And because of cross-linking three-dimensional network which is formed, solidification happens, and what you get is a thermoset solid polymer, and that is how the two surfaces got bound together.
- Addition one is the low molecular weight polymer, which is soluble, and the other one, which we usually refer to as hardener, is actually known as a curing agent, and that is responsible for the solidification of the.
- Polymer and that happens due to three-dimensional cross-linking network getting form.
- There are certain materials for which you have to use a particular type of adhesive.
- For wood, it's usually urea formalhydrasin, but epoxy resin is a very good resin because.
- because it can be used to attach metal, wood, glass, concrete, leather, any surfaces, epoxy resin can act as a very good adhesive.
- What is an epoxy group?
- It is a three-membered ring.
- of an O atom bonded to carbon atoms.

- Three member rings are usually very reactive.
- Now, bisphenol A is the first monomer which is being used.
- When these react, what we get is the bisphenol A.
- The other monomer is epi chloroth.
- We add Cl2, what we get is the allyl chloride.
- That is now treated with HOCl.
- Dehydration is going to happen.
- It's sorry, it's Cl will be removed, and you get a three-membered ring over there.
- So this is known as Epichlorohydrine and you get NaCl because you are treating with it NaOH plus this HCl so you get the NaCl.

Example or application from the lecture

- Let us first define adhesives, yes, substances which are used to bind two or more materials firmly by surface attachment due to the penetration of the adhesive into the material and solidification which leads to interfacial locking.
- For example, for rubber we use phenol formaldehydrin.
- Let us see the synthesis.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000098 · EV0000099 · EV0000100 · +1 source units in pack.json

Core explanation · 05:00–09:59

And what do we notice here that there is an OH group over here and a Cl.

Key details

- These two molecules can undergo condensation, and this would happen in the presence of NaOH.
- HCl is getting eliminated.
- And what would happen as a result of this condensation?
- We can see here that there is an attachment between the bisphenol A and.
- And the epi chlorhydrate.
- This ring has opened up because if you look at this monoma.
- There are two hydroxyl groups on both sides, therefore it would classify as a monomalase.
- In the first class I mentioned, at least two functional groups should be there.
- But in this epiclorohydrate, we find there is just one CL which can take part in the condensation, whereas here we don't find anything, but yes.
- That is the reason why we have the ring here.
- Now this ring being very reactive, it is going to open up when this condensation will happen and therefore this ring is going to open up like this and we will find that we get the ch oh you can see here the ch oh is formed and now this ch2 is free to condense with the next bisphenol A ring.
- So here now this we have Cl condensation on one side and the other side it is CH2 where the ring is opened up and there's a free bond here that is going to react with the next bisphenol A.

- So this is the mechanism, and therefore we get this as the main repeat unit in the epoxy polymer.
- Epoxy resin is formed, an excess of epichlorohydrine is used.
- What would be the reason for that?
- The reason is that the N groups we want them to be epoxy groups.
- make sure that the that happens by using an excess of epi chlorohyain.
- When we use excess of epichlorohyden, because there is no more bisphenol, further reaction with bisphenole is not possible.
- If there was more bisphenol as we understood the mechanism just now, this ring would open.
- We'll get a CHOH and the CH2 combined would have combined with another bisphenole, but there is no bisphenol, so the ring remains intact, and in the end, we have ensured that both the side chains, the end of terminal chains, terminal groups are epoxy groups.
- That because we know that epoxy is a very reactive group, and when we add the curing agent, immediately there will be a.
- Three dimensional cross link structure which is formed, and we know that when between these two chains we have cross-link structure getting formed, the polymer is going to solidify.
- Use we use diamines, di carboxylic acid, or acid anhydrides.
- Because there are these reactive epoxy groups, immediately cross link structure will be formed, which will result in solidification, and hence the two surfaces are going to get bound.
- What are the properties of these epoxy resins?
- They have, of course, excellent addition, excellent resistance to water, acid, alkali, and other corrosive chemicals.
- And they also have very high mechanical strength.
- They absorb less moisture.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000102 · EV0000103 · EV0000104 · +1 source units in pack.json

Core explanation · 10:00–14:59

and they have very good insulation properties.

Key details

- They are also used to bind glass, metal, leather, wood, ceramic, etc.
- Of course, that is the reason why they come under the class of adhesives.
- Then when they are coated on the floor, they give and anti-skid layer.
- They are used for laminating materials and for crease and shrinkage resistance in fabric.
- So in the next class we are going to talk of another important class of polymers or rather polymer composites.
- So we will take up Kepler in the next class.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000106 · EV0000107 · EV0000108 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Synthesis and properties of epoxy resin transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/045_Synthesis_and_properties_of_epoxy_resin__7c6d59e1128dc6e8.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Synthesis and properties of epoxy resin transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/045_Synthesis_and_properties_of_epoxy_resin__7c6d59e1128dc6e8.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Synthesis and properties of epoxy resin transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/045_Synthesis_and_properties_of_epoxy_resin__7c6d59e1128dc6e8.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Synthesis and properties of epoxy resin transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_04/AV_Summaries/045_Synthesis_and_properties_of_epoxy_resin__7c6d59e1128dc6e8.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.

4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00022	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0046. Sidecar variants compare...
SRC00023	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0046. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00024	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0046. Sidecar variants compare...
SRC00025	3 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0022. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0023 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0046. Sidecar variants compare...

C Revision and retrieval practice